

Spectrogram and Audio Features

From STFT to Mel spectrogram to MFCCs — the feature representations that power modern speech AI

9 min

Duration

B3 – Apply

Bloom's

L2 – Intermediate

Level

TH-000006

Prerequisite

Session Overview

What you will learn and the conceptual roadmap for this session

What You Will Learn

- What a spectrogram is and how it is computed
- The Short-Time Fourier Transform (STFT) formula and its parameters
- Mel spectrogram: why perceptual scaling improves speech AI
- MFCCs and their Discrete Cosine Transform step
- Delta and delta-delta features for capturing temporal dynamics
- How to choose the right feature for your task

Concept at a Glance

- Duration: 9 minutes
- Bloom's Level: B3 – Apply
- Difficulty: L2 – Intermediate
- Module: 4.2 Audio and Speech Models
- Prerequisite: Audio Processing Fundamentals (TH-000006)
- Next: ASR Models (TH-000008)

Learning Objectives

By the end of this session you will be able to:

9 min

Duration

B3 – Apply

Bloom's

L2 – Intermediate

Level

TH-000006

Prerequisite

- 1 Explain how the Short-Time Fourier Transform decomposes audio into time-frequency representations
- 2 Compute a spectrogram from a raw waveform conceptually, identifying all key parameters
- 3 Construct the Mel spectrogram pipeline and explain why perceptual scaling improves speech AI
- 4 Apply the MFCC computation steps and identify when MFCCs are preferred over Mel spectrograms
- 5 Select the appropriate audio feature representation for a given AI application

SECTION 1 — WHY TIME-FREQUENCY ANALYSIS

The Problem with Time-Domain and Frequency-Domain Alone

A raw waveform or a plain FFT are each missing half the picture

Why Neither Time-Domain Nor Frequency-Domain Is Enough

Speech, music, and environmental sounds are non-stationary — their frequency content changes over time

Time Domain (Raw Waveform)

✓ Shows when events happen (clicks, silence, onsets)

✗ Cannot tell you **WHAT** frequencies are present at each moment. The /s/ in 'speech' and the /ee/ vowel look very similar as amplitude values.

Frequency Domain (Full FFT)

✓ Shows all frequencies present in the signal

✗ Time information is completely collapsed — you cannot tell **WHEN** each frequency occurred. Useless for speech recognition.

The Solution: Short-Time Fourier Transform (STFT)

Apply the Fourier Transform to short overlapping windows of the signal. This gives us a 2D time-frequency representation: horizontal axis = time, vertical axis = frequency, pixel intensity = energy. Models can 'see' speech the way humans intuitively hear it — as an image.

SECTION 2 — THE STFT AND SPECTROGRAM

The Short-Time Fourier Transform

The mathematical engine behind every spectrogram in audio AI

The STFT Formula

The mathematical definition — each term has a precise physical meaning

$$\text{STFT}\{x\}(m, k) = \sum_{n=0}^{N-1} x(n+mH) \cdot w(n) \cdot e^{(-j2\pi kn/N)}$$

m = frame index | k = frequency bin index | H = hop length | $w(n)$ = window function | N = FFT size

$x(n+mH)$:

The audio sample at position n within frame m , offset by hop H . This is the raw signal being analysed.

$w(n)$:

The window function (Hann/Hamming). Tapers the frame to zero at its edges to prevent spectral leakage.

$e^{(-j2\pi kn/N)}$:

Complex sinusoid at frequency bin k . The Fourier basis function. Rotates in the complex plane at frequency k/N cycles per sample.

$|\text{STFT}|^2$ or $|\text{STFT}|$:

Take squared magnitude for power spectrogram, magnitude for amplitude spectrogram. Phase information is discarded for most AI applications.

Output shape: For a 5-second audio at 16 kHz with FFT size 512, hop 160: output is (257, 501) — 257 frequency bins $\times$ 501 time frames. Spectrogram = 257 magnitude values per frame stacked into a 2D matrix.

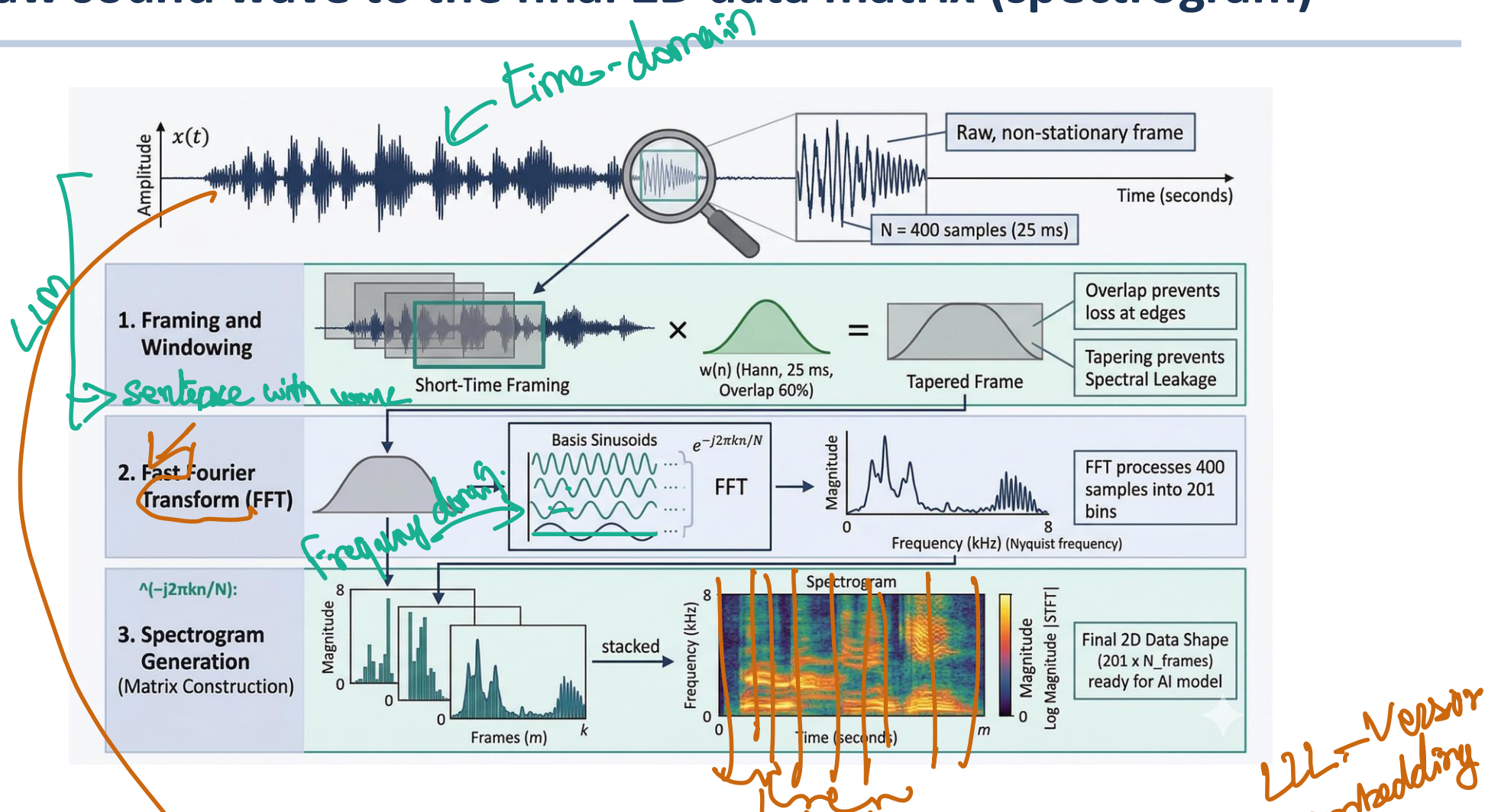
STFT Parameters and Their Effects

Every parameter affects the time-frequency resolution trade-off — there is no free lunch

Parameter	Typical Value	Whisper Value	Effect of Increasing
FFT size (N)	512	512	More frequency resolution, wider time window, more smearing
Frame length	400 samples (25 ms)	400 samples	Longer: better freq resolution; worse time resolution
Hop length (H)	160 samples (10 ms)	160 samples	Smaller hop: more frames, finer time resolution, more compute
Window function	Hann	Hann	Different windows trade leakage for resolution differently
Output shape	$(N/2+1) \times T$	(257, 3000)	For 30-sec audio: 3000 time frames, 257 frequency bins

Heisenberg trade-off: You cannot simultaneously achieve perfect time resolution AND perfect frequency resolution. A longer frame gives better frequency resolution but blurs time. Short frames give sharp time resolution but poor frequency resolution. 25 ms is the empirical optimum for speech.

Raw sound wave to the final 2D data matrix (spectrogram)



SECTION 3 — MEL SPECTROGRAM

The Mel Spectrogram

Perceptual scaling: compressing 257 linear bins into 80 Mel-spaced bins that match how humans hear



The Mel Scale: Perceptual Frequency Mapping

Humans perceive pitch logarithmically — the Mel scale captures this non-linear relationship

$$m = 2595 \cdot \log_{10}(1 + f / 700) \quad \Leftrightarrow \quad f = 700 \cdot (10^{(m/2595)} - 1)$$

m = Mel frequency | f = linear frequency in Hz | 700 Hz = the perceptual crossover point

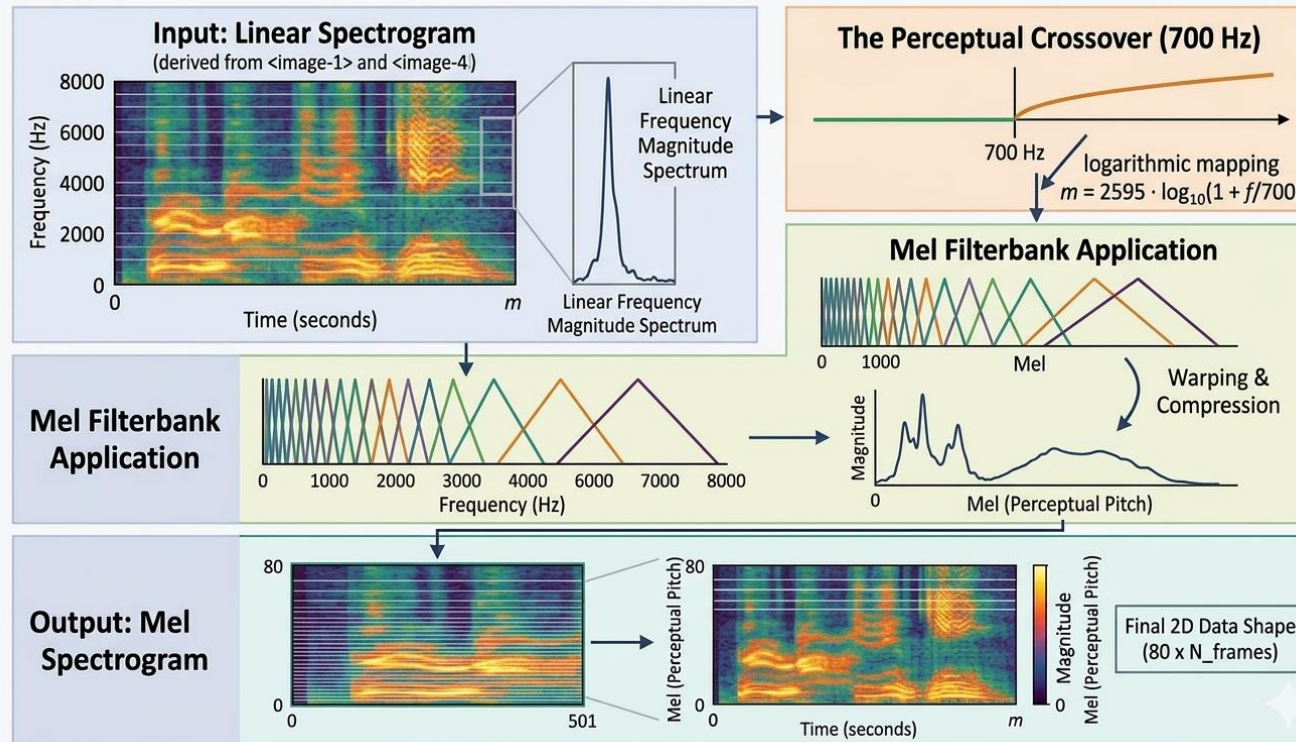
Why the Mel Scale Matters for AI

- Human hearing discriminates small differences in low frequencies (100 Hz vs 200 Hz) but cannot distinguish large differences in high frequencies (8,000 Hz vs 8,100 Hz).
- The Mel filterbank implements this by placing narrow triangular filters at low frequencies (fine resolution) and wide filters at high frequencies (coarse resolution).
- This compression from 257 linear bins to 80 Mel bins is NOT a loss of information — it discards frequency detail humans cannot perceive anyway.
- Result: the Mel spectrogram is a more compact, more perceptually relevant representation. Neural networks converge faster and generalise better on Mel features than linear spectrograms.

Intuition: Between 100 Hz and 200 Hz (100 Hz apart), you hear a full octave — a HUGE perceptual difference. Between 8,000 Hz and 8,100 Hz (also 100 Hz apart), you hear almost nothing. The Mel scale encodes this perceptual non-linearity.

The Mel Scale: Perceptual Frequency Mapping

Humans perceive pitch logarithmically — the Mel scale captures this non-linear relationship



From STFT to Log-Mel Spectrogram

The four-step pipeline that transforms a raw spectrogram into the de-facto standard AI audio feature

STFT Magnitude

257 frequency bins
complex $\rightarrow |X|$

Mel Filterbank

80 triangular filters
(or 128 for music)

Log Compression

$\log(1 + \text{mel_energy})$
mimics loudness

Log-Mel Spectrogram

Shape: $80 \times T$
ready for model

Whisper specifics: 80 Mel bins, 25 ms window (400 samples), 10 ms hop (160 samples), 512-point FFT, $f_{\min}=0$ Hz, $f_{\max}=8000$ Hz. For 30-second chunks: output shape = (80, 3000). The log-Mel spectrogram IS the model input — changing it invalidates pretrained weights.

SECTION 4 — MFCCS

Mel-Frequency Cepstral Coefficients (MFCCs)

Adding a DCT step to decorrelate Mel features — the legacy standard, still relevant for edge AI

MFCC Computation Pipeline

Six steps from raw audio to the 13-39 dimensional feature vectors used by HMM-based ASR and edge systems

1**Pre-emphasis**

$y[n] = x[n] - 0.97 \cdot x[n-1]$
Boosts high-freq content before analysis

2**Framing & Windowing**

25 ms frames, 10 ms hop
Hann or Hamming window applied

3**FFT**

N-point DFT per frame
Output: complex magnitude spectrum

4**Mel Filterbank**

80 triangular filters applied
Compress to Mel-spaced energy bins

5**Log Compression**

Take log of filter bank energies
Mimics human loudness perception

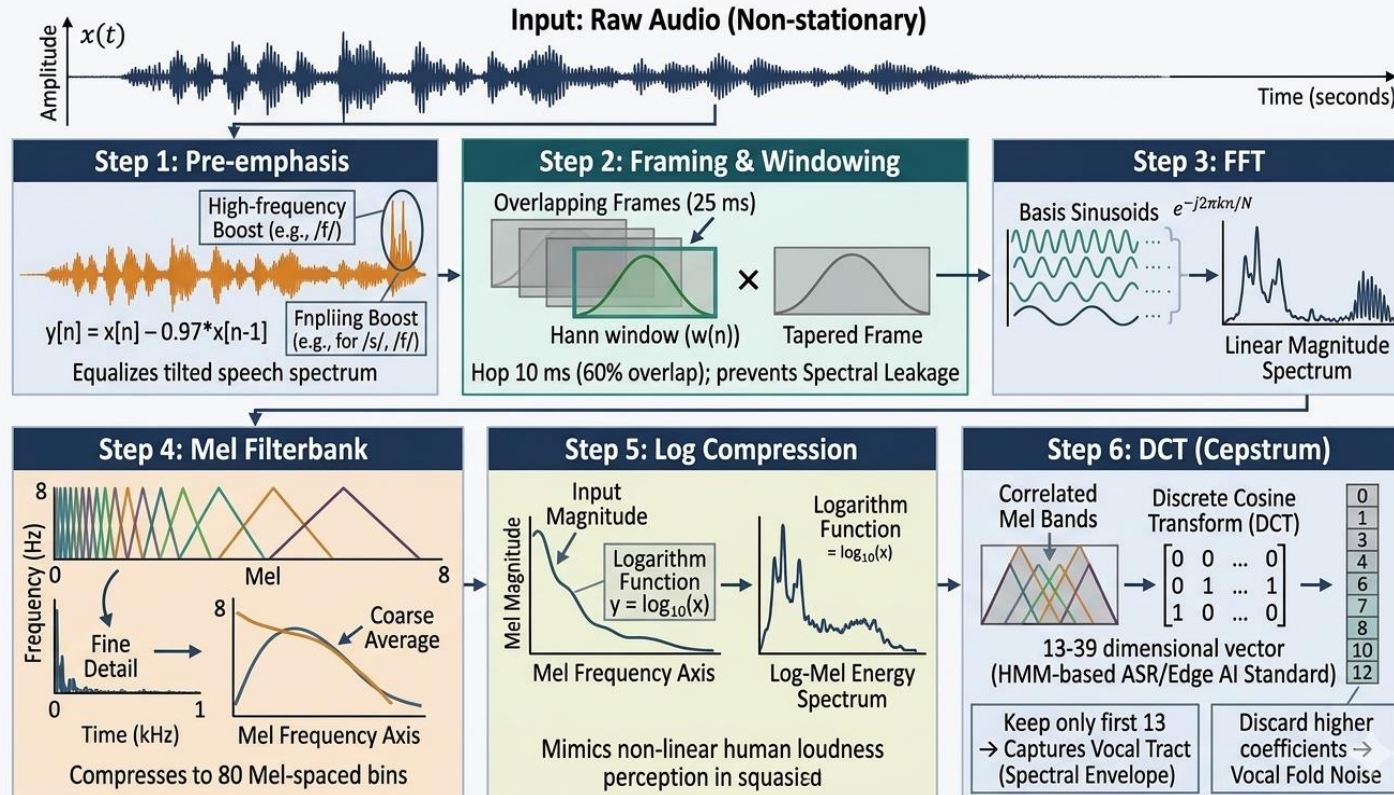
6**DCT (Cepstrum)**

Discrete Cosine Transform
Decorrelates Mel bands → 13 coefficients

Why DCT? Adjacent Mel filterbank channels overlap heavily, making their outputs correlated. The DCT decorrelates them — like PCA. Keeping only the first 13 coefficients captures the spectral envelope (vowel formants) while discarding fine spectral detail (noise).

MFCC Computation Pipeline

Six steps from raw audio to the 13-39 dimensional feature vectors used by HMM-based ASR and edge systems



Audio Feature Comparison: Which to Use When

Choose your feature representation based on your task, compute budget, and model architecture

Feature	Dim/Frame	Compute	Best For	Used In
Raw Waveform	16,000/sec	Low	End-to-end self-supervised	wav2vec 2.0, HuBERT
STFT Magnitude	257	Low	Reconstruction, vocoders	HiFi-GAN, WaveNet input
Log Spectrogram	257	Low	CNN classifiers	Audio event detection
Log-Mel Spectrogram	80–128	Low	Speech AI (current standard)	Whisper, Tacotron 2
MFCC	13–40	Low	Edge/embedded ASR	Kaldi, iOS SpeechKit
MFCC + Δ + $\Delta\Delta$	39	Medium	Classic HMM-ASR	Legacy enterprise ASR

Field consensus: The field has converged on Log-Mel Spectrogram (80 bins) as the standard input for deep learning speech models. MFCCs are used for edge/embedded applications where memory is critical. Leading SOTA (wav2vec, HuBERT) learns representations directly from raw waveforms.

SECTION 5 — GENAI APPLICATIONS

Spectrograms in Production AI

How these features power Whisper, TTS systems, and modern GenAI audio pipelines

Whisper and Modern ASR: How Spectrograms Are Used

Whisper's audio processing pipeline is the industry template for large-scale speech recognition

Whisper's Audio Processing (OpenAI, 2022)

Input Any audio format → resample to 16 kHz mono internally

Chunking Split into 30-second segments with 50% overlap

Log-Mel 80 bins, 25 ms window, 10 ms hop → padded to 3,000 frames

Encoder 2×Conv1D (stride 1,2) reduce 3,000 → 1,500 frames → 6 Transformer layers

Decoder Autoregressive Transformer conditioned on audio via cross-attention

Multitask Same architecture: transcribe, translate, identify language

SpecAugment: The Standard Augmentation

What it is Augment the Mel spectrogram directly — simpler and more effective than raw audio augmentation

Frequency masking Randomly zero out F consecutive Mel bins → model robust to missing frequencies

Time masking Randomly zero out T consecutive time steps → model uses broader context

Result SpecAugment reduced LibriSpeech WER from 6.8% to 5.8% (best 2019) with no additional data

Used by Conformer, wav2vec 2.0, Whisper fine-tuning, virtually all modern ASR training

Engineer takeaway: The Mel spectrogram is part of the model contract — changing the number of Mel bins, window size, or hop length invalidates pretrained weights. Always match the preprocessing exactly to the model's training configuration.

Spectrograms in TTS: The Intermediate Representation

In generation, Mel spectrograms bridge text-to-acoustic models and waveform synthesis

Tacotron 2 (Google, 2018)

Text → attention seq2seq → 80-bin Mel spectrogram target → WaveNet vocoder converts Mel to waveform. The Mel spectrogram is the explicit intermediate representation connecting the acoustic model to the vocoder.

FastSpeech 2 (Microsoft, 2020)

Predicts 80-bin Mel spectrogram from phonemes with explicit duration, pitch (F0), and energy predictors. Log-Mel spectrogram is the direct training target. Control via: Mel duration scaling, F0 offset, energy scaling.

VITS (2021)

Eliminates the two-stage pipeline. Still internally uses a posterior encoder that operates on spectrogram-like representations. A normalising flow learns the mapping from latent to Mel directly.

VALL-E (Microsoft, 2023)

Replaces Mel spectrograms entirely — uses EnCodec discrete tokens (8 codebooks, 75 tokens/sec). An LLM generates codec tokens. Codec decoder converts back. Mel spectrograms are no longer the intermediate for token-based TTS.

Trend: Discrete neural audio codecs (EnCodec, SoundStream, DAC) are replacing Mel spectrograms in generative models. Mel spectrograms remain the standard for analysis and ASR. Both representations coexist.

Key Takeaways

The six essential points from this concept

STFT decomposes audio into time-frequency Apply windowed FFT to overlapping 25 ms frames. Output: complex matrix (257 freq bins $\times$ T frames).
Spectrogram = $|\text{STFT}|^2$.

Mel filterbank compresses to perceptual scale 257 linear bins $\rightarrow$ 80 Mel bins via triangular filters. Log compression $\rightarrow$ log-Mel spectrogram. This is the de-facto standard for speech AI.

MFCCs add a decorrelation step Mel energies $\rightarrow$ log $\rightarrow$ DCT $\rightarrow$ 13 coefficients. Used for edge/embedded systems where memory is critical. Static MFCCs + Δ + $\Delta\Delta$ = 39-dim vector per frame.

SpecAugment is the standard augmentation Mask random frequency and time bands directly on the Mel spectrogram. Reduces WER significantly. Used in all modern ASR training pipelines.

Whisper sets the industry template 80 Mel bins, 25 ms/10 ms, 30-sec chunks. The Mel spectrogram IS the model contract — match it exactly during inference.

Note: Next: AI-MM-IN-TH-000008 — ASR Models (Conformer, Whisper architecture, CTC, wav2vec 2.0, WER)

AI-MM-IN-TH-000007 Complete

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